Name: _____ **Roll number:** _____

Part A: predicting a number (regression), worked example

Self-paced practice: nothing to hand in, not graded, not part of any quiz. Read this side first, then try side 2.

The setting. A model answers requests. For four requests we know how long the prompt was, x (in hundreds of tokens), and how long the model took to answer, y (in tens of milliseconds). We want to predict y from x. XGBoost begins with one guess for every request. Then it adds small trees, one after another, and each tree corrects part of what the guess still gets wrong. It has three settings: $\lambda = 1$ (a brake on big steps), $\gamma = 1$ (the price of making a split) and $\eta = \frac{1}{2}$ (the fraction of each tree we add).

request i	1	2	3	4
x_i	1	2	3	4
y_i	3	5	9	11

Step 1 **Start with one guess for every request.**

We know nothing yet, so the best single guess is the average of y: $F_0 = (3+5+9+11)/4 = \mathbf{7}$.

Step 2 **Which way should each guess move?**

We measure a mistake by the squared error $\frac{1}{2}(F-y)^2$ (the $\frac{1}{2}$ keeps the next formulas short). Ask: if we raise the guess a little, does the mistake grow or shrink, and how fast? That rate is the **slope**, $g = F - y$. Then ask: how quickly does the slope itself change? That is the **curvature**, h. Here it is $h = 1$. A positive slope means the guess is too high, so it should come down. A negative slope means it is too low.

request i	1	2	3	4
slope $g_i = 7 - y_i$	4	2	-2	-4
curvature h_i	1	1	1	1

Step 3 **What number should a leaf add?**

A tree sorts the requests into groups called **leaves**. Every request in a leaf gets the same number w added to its guess. For the requests in one leaf, add up the slopes, $G = \sum g_i$, and add up the curvatures, $H = \sum h_i$. If we add a small w, the loss changes by about $G\,w$ (slope times step), plus $\frac{1}{2}H\,w^2$ (the loss bends upward). We also add a penalty $\frac{1}{2}\lambda w^2$, so that w cannot grow too large. Together:

$$Q(w) = Gw + \tfrac{1}{2}(H+\lambda)w^2.$$

The best w is the lowest point of Q, where its slope is zero: $G + (H+\lambda)w = 0$, so

$$\boxed{w = -\frac{G}{H+\lambda}}$$

A bigger λ makes the number we divide by bigger, so w gets smaller: that is the brake. Now put this w back into Q: $Q = -\frac{G^2}{H+\lambda} + \frac{1}{2}\frac{G^2}{H+\lambda} = -\frac{1}{2}\frac{G^2}{H+\lambda}$. So the leaf saves $\frac{1}{2}\,G^2/(H+\lambda)$ of loss. We call $G^2/(H+\lambda)$ the **leaf score**: the bigger it is, the more the leaf saves.

Step 4 **How good is a split?**

A split "$x \le t$" sends the requests with $x \le t$ to the left leaf and the rest to the right leaf. With all four requests in one leaf, $G = 0$ and $H = 4$, so the score is $0^2/(4+1) = 0$. After a split, each of the two leaves has its own score. The **gain** tells us how much the split saves:

$$\text{gain} = \tfrac{1}{2}\big[\text{score}_L + \text{score}_R - \text{score}_{\text{all}}\big] - \gamma.$$

The γ is the price of making a split, so a split is worth making only if its gain is above 0. We try all three splits:

split	G_L	H_L	G_R	H_R	$\frac{G_L^2}{H_L+\lambda}$	$\frac{G_R^2}{H_R+\lambda}$	gain
$x \le 1$	4	1	-4	3	8	4	5
$x \le 2$	6	2	-6	2	12	12	11
$x \le 3$	4	3	-4	1	4	8	5

Look at $x \le 2$: $G_L = 4+2 = 6$ and $H_L = 2$, so the left score is $\frac{36}{2+1} = 12$. The right side has $G_R = -6$, $H_R = 2$, so its score is also 12. The gain is $\frac{1}{2}(12+12-0) - 1 = 11$. This is the largest gain, so the best split is $x \le 2$. It keeps the two requests whose guess is too high together, and the two whose guess is too low together.

Step 5 **Leaf values.**

Left leaf: $w_L = -6/(2+1) = \mathbf{-2}$. Right leaf: $w_R = \mathbf{+2}$. The left leaf pulls its guesses down: their slopes were positive.

Step 6 **Update the guess.**

We add only a part of w: $F_1 = F_0 + \eta\,w$ with $\eta = \frac{1}{2}$. Left requests: $7 + \frac{1}{2}(-2) = 6$. Right requests: $7 + \frac{1}{2}(2) = 8$.

request i	1	2	3	4
y_i	3	5	9	11
new guess F_1	6	6	8	8

Total squared error $\sum(y_i - F_i)^2$. With F_0: $16+4+4+16 = 40$. With F_1: $9+1+1+9 = \mathbf{20}$. The error halved. We add only half of w so one rough tree cannot push the guesses too far.

Step 7 **Repeat.**

The new slopes are $g_i = F_1 - y_i = 3,\ 1,\ -1,\ -3$, half of the old ones. The next tree starts from these slopes and goes back to Step 3. **That is all XGBoost does, again and again:** work out the slopes, pick the split with the best gain, set the leaf values, and add a part of them.

Step 8 **The two settings.**

λ **is a brake.** Left leaf: $w_L = -6/(2+\lambda)$ is -3 for $\lambda = 0$, -2 for $\lambda = 1$, -1 for $\lambda = 4$: a bigger λ gives smaller steps. γ **is the price of a split.** Best gain before the price: 12. With $\gamma = 1$ it is 11: we split. With $\gamma = 15$ it is -3: no split.

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Part A: predicting a number (regression), now you do it

Same steps as side 1, on new data. Where you see a), b): circle the one right answer. Write your working in the boxes.

request i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	3	5	4	10	12	11

Same settings: $\lambda = 1$, $\gamma = 1$, $\eta = \frac{1}{2}$. Slope $g = F - y$, curvature $h = 1$, $w = -G/(H+\lambda)$, score $= G^2/(H+\lambda)$, gain $= \frac{1}{2}[\text{score}_L + \text{score}_R - \text{score}_{\text{all}}] - \gamma$, $F_1 = F_0 + \eta w$.

Step 1 **First guess, slopes, curvatures.**

$F_0 =$ _____

request i	1	2	3	4	5	6
slope $g_i = F_0 - y_i$						
curvature h_i						

Step 2 **Try three splits.**

(All six together: $G =$ _____, $H =$ _____, score = _____.)

split	G_L	H_L	G_R	H_R	$\frac{G_L^2}{H_L+\lambda}$	$\frac{G_R^2}{H_R+\lambda}$	gain
$x \le 2$							
$x \le 3$							
$x \le 4$							

The best split is **a)** $x \le 2$ **b)** $x \le 3$ **c)** $x \le 4$ because **a)** it has the largest gain **b)** it has the smallest gain.

Step 3 **Leaf values.**

Left leaf $G_L =$ _____, $H_L =$ _____, $w_L =$ _____ Right leaf $G_R =$ _____, $H_R =$ _____, $w_R =$ _____

Step 4 **Update the guess, then the new slopes.**

request i	1	2	3	4	5	6
new guess F_1						
new slope g_i						

The total squared error was 77.5 with F_0. With F_1 it is **a)** 32.7 **b)** 52.7 **c)** 77.5. The new slopes are **a)** smaller than **b)** the same as **c)** larger than the old ones, so the next tree has **a)** less **b)** more to fix.

Step 5 **The two settings.**

(a) Left leaf with $G_L = 10.5$, $H_L = 3$: $w_L = -G_L/(H_L + \lambda)$.

λ	0	1	5
w_L			

A larger λ gives **a)** smaller steps **b)** larger steps

(b) Before the price the best gain is $\frac{1}{2}(55.125 - 0) = 27.5625$. With $\gamma = 30$ the gain is _____ so XGBoost **a)** makes the split **b)** does not make the split.

In your own words: what does XGBoost do in one round?

Name: _____ **Roll number:** _____

Part B: predicting yes or no (classification), worked example

Read this side first. It is the same loop as Part A; only the slope and the curvature change. On side 4 you do it yourself.

The setting. A model flags emails as spam ($y = 1$) or not spam ($y = 0$). For four emails we know x, the number of suspicious words, and y. The model now gives each email a **score** F, which can be any number. A big positive score means "probably spam". The score is turned into a chance of spam by $p = \dfrac{1}{1+e^{-F}}$; for example $F = 0$ gives $p = 0.5$. Settings: $\lambda = 1$, $\gamma = 0$ (no price for a split this time), $\eta = \frac{1}{2}$.

email i	1	2	3	4
x_i	1	2	3	4
y_i	0	0	1	1

Step 1 **Start with one score for everybody.**

Half of the emails are spam, so the best single chance is 0.5. To go back from a chance to a score we use $F = \ln \dfrac{p}{1-p}$. For $p = 0.5$ this is $F_0 = \ln 1 = \mathbf{0}$. (If 80% were spam: $F_0 = \ln \dfrac{0.8}{0.2} = \ln 4 = 1.386$.)

Step 2 **Slope and curvature for a yes/no answer.**

If $y = 1$ we want p close to 1, and if $y = 0$ we want p close to 0. The **log loss** measures this: $\ell = -\big[y \ln p + (1-y)\ln(1-p)\big]$. To find its slope we use $\dfrac{dp}{dF} = p(1-p)$. Then the slope is $g = \dfrac{d\ell}{dF} = \Big[-\dfrac{y}{p} + \dfrac{1-y}{1-p}\Big] p(1-p) = \boxed{p - y}$ and the curvature is $h = \dfrac{dg}{dF} = \boxed{p(1-p)}$. As before, a positive g means the chance is too high, so the score should come down.

email i	1	2	3	4
slope $g_i = 0.5 - y_i$	0.5	0.5	−0.5	−0.5
curvature $h_i = 0.5 \times 0.5$	0.25	0.25	0.25	0.25

Step 3 **Leaf value and score: nothing new.**

A leaf adds the same w to the score of every email in it. With $G = \sum g_i$ and $H = \sum h_i$ for the leaf, the lowest point of $Q(w) = Gw + \frac{1}{2}(H+\lambda)w^2$ is again $w = -\dfrac{G}{H+\lambda}$, and the leaf score is $\dfrac{G^2}{H+\lambda}$. Here w is a change in the *score* F, not in the chance.

Step 4 **Try the three splits.**

gain $= \frac{1}{2}[\text{score}_L + \text{score}_R - \text{score}_{\text{all}}] - \gamma$. All four emails: $G = 0$, $H = 1$, score 0.

split	G_L	H_L	G_R	H_R	$\dfrac{G_L^2}{H_L+\lambda}$	$\dfrac{G_R^2}{H_R+\lambda}$	gain
$x \le 1$	0.5	0.25	−0.5	0.75	0.2	0.143	0.171
$x \le 2$	1	0.5	−1	0.5	0.667	0.667	0.667
$x \le 3$	0.5	0.75	−0.5	0.25	0.143	0.2	0.171

For $x \le 2$: $\frac{1^2}{0.5+1} = 0.667$ on each side, so the gain is $\frac{1}{2}(0.667 + 0.667 - 0) = 0.667$. The best split is $x \le 2$: it keeps the two non-spam emails together and the two spam emails together.

Step 5 **Leaf values.**

$w_L = -1/(0.5+1) = \mathbf{-0.667}$, $w_R = \mathbf{+0.667}$. The left leaf lowers the spam score, because its emails are not spam.

Step 6 **Update the score, then the chance.**

$F_1 = F_0 + \eta w$: left $0 + \frac{1}{2}(-0.667) = -0.333$, right $+0.333$. The chance is $p = 1/(1+e^{-F})$: for $F = 0.333$, $e^{-0.333} = 0.717$, so $p = 1/1.717 = 0.583$; for $F = -0.333$, $p = 1 - 0.583 = 0.417$.

email i	1	2	3	4
y_i	0	0	1	1
new score F_1	−0.333	−0.333	0.333	0.333
new chance p_1	0.417	0.417	0.583	0.583

Chance of spam went down for non-spam and up for spam: the right way. Total log loss $\sum \ell$: from $4 \ln 2 = 2.773$ to 2.161.

Step 7 **Repeat.**

New slopes $g_i = p_1 - y_i = 0.417, 0.417, -0.417, -0.417$: smaller than before (0.5). The next tree starts from them. The loop is the same as in Part A: slopes and curvatures, best split, leaf values, add a part.

Step 8 **What changed from Part A?**

Only three things: the loss is the log loss; $g = p - y$ and $h = p(1-p)$ replace $g = F - y$ and $h = 1$; and the score F is turned into a chance by $p = 1/(1+e^{-F})$. The leaf rule $w = -G/(H+\lambda)$, the gain, λ, γ and η work exactly as before.

Name: _____ **Roll number:** _____

Part B: predicting yes or no (classification), now you do it

Same steps as side 3, on new data. Where you see a), b): circle the one right answer. Write your working in the boxes.

email i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	0	0	0	1	1	1

Settings: $\lambda = 1$, $\gamma = 0$, $\eta = \frac{1}{2}$. $p = 1/(1+e^{-F})$, slope $g = p - y$, curvature $h = p(1-p)$, $w = -G/(H+\lambda)$, gain $= \frac{1}{2}[\text{score}_L + \text{score}_R - \text{score}_{\text{all}}]$, $F_1 = F_0 + \eta w$.

Step 1 **First score, slopes, curvatures.**

Half of the emails are spam, so $F_0 =$ _____ and $p_0 =$ _____. (If 90% were spam, $F_0 = \ln \frac{0.9}{0.1} =$ _____.)

email i	1	2	3	4	5	6
slope $g_i = p_0 - y_i$						
curvature h_i						

Step 2 **Try three splits.**

(All six together: $G =$ _____, $H =$ _____, score = _____.)

split	G_L	H_L	G_R	H_R	$\frac{G_L^2}{H_L+\lambda}$	$\frac{G_R^2}{H_R+\lambda}$	gain
$x \le 2$							
$x \le 3$							
$x \le 4$							

The best split is **a)** $x \le 2$ **b)** $x \le 3$ **c)** $x \le 4$ because **a)** it separates the spam from the non-spam **b)** it splits the emails evenly.

Step 3 **Leaf values.**

Left leaf $G_L =$ _____, $H_L =$ _____, $w_L =$ _____Right leaf $G_R =$ _____, $H_R =$ _____, $w_R =$ _____

Step 4 **Update the score, the chance, and the new slopes.**

$F_1 = F_0 + \eta w$, then $p_1 = 1/(1+e^{-F_1})$ (use a calculator), then $g = p_1 - y$.

email i	1	2	3	4	5	6
new score F_1						
new chance p_1						
new slope g_i						

For the non-spam emails the chance of spam went **a)** down **b)** up; for the spam emails it went **a)** down **b)** up. The new slopes are **a)** smaller than **b)** the same as **c)** larger than the old ones.

Step 5 **Compare with Part A.**

(a) The slope for a yes/no problem is **a)** $F - y$ **b)** $p - y$ and the curvature is **a)** 1 **b)** $p(1-p)$.

(b) An email with $y = 1$ already has $p = 0.99$. Its slope $g =$ _____ and curvature $h =$ _____. It is almost right, so it has **a)** a big effect **b)** very little effect on the next tree.

In your own words: what is different about the yes/no case, and what stays the same?